Lab 3 — Mixing and Balancing

Friday, October 30 (~20 min)

Last updated: October 12, 2026.

In class Friday, October 30 (same day as Lecture 29 / Quiz 8). Quiz 8 is Week 9 dynamics — **not** this lab.

Credit: Lab points are for effort in class, not for a Canvas file upload. Preview the one-pager; write on the hard copy handed out in class.

Learning objectives

1. Write a small contextual system as $Ax = b$.
2. Solve with software and check the residual $Ax - b$.
3. Prefer solving over forming A^{-1} when the goal is x.

Math connections

Systems and RREF (Lectures 21–23); later you will connect this to inverses (Lecture 30).

Key syntax (this lab)

Task	Octave	Python
Solve $Ax = b$	`x = A\b`	`x = np.linalg.solve(A,b)`
Residual	`A*x - b`	`A @ x - b`
Inverse (avoid for solving)	`inv(A)`	`np.linalg.inv(A)`

Ops strip

Café: tea \$2, coffee \$3. Sold x_1 teas and x_2 coffees; revenue \$26; 10 drinks total.

C3.0. Write A, x, and b for the two equations on paper. Enter A and b in software. Print `size(A)` / `A.shape`.

Expected:

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 26 \\ 10 \end{bmatrix}$$

(A is 2×2).

Application

C3.1. Solve $Ax = b$. Report x.

Expected: $x = (4, 6)$ (4 teas, 6 coffees).

C3.2. Compute the residual $Ax - b$. Is it (numerically) zero?

Expected: $(0, 0)$ (or $\sim 10^{-15}$).

C3.3. True or false, with one line: to solve $Ax = b$ you should usually compute $x = A^{-1}b$ explicitly.
Then time (roughly) `A\b` vs `inv(A)*b` / `inv(A) @ b` once — which felt simpler?

Expected: False — prefer a solver; inverse is extra work and can be less stable.

One-pager

Preview the Lab 3 ticket; write on the hard copy in class.